TEXORA · LECTURE SHEET

Moisture and Count Calculations

Textiles are bought and sold by weight, but part of that weight is water — and the amount changes with the weather. This sheet covers regain and moisture content, correction of count to standard regain, invoice weight, count-system conversions, and count from a weighed lea.

YM-3207 · ADVANCED SPINNING CALCULATIONS · TOPIC 6 · SPECIALIZATION — YARN MANUFACTURING

1. Regain and moisture content — two different fractions

A textile sample consists of oven-dry fibre plus absorbed water. The water can be expressed on either basis, and confusing the two is the most common error in this topic.

$$\text{Regain (\%)} = \text{mass of water} / \text{oven-dry mass} \times 100$$

$$\text{Moisture content (\%)} = \text{mass of water} / \text{total (conditioned) mass} \times 100$$

WORKED EXAMPLE — REGAIN AND MOISTURE CONTENT OF ONE SAMPLE

A conditioned sample weighs 105.0 g; after oven drying it weighs 97.2 g.

Water = 105.0 − 97.2 = 7.8 g.

Regain = 7.8 / 97.2 × 100 = **8.02%**.

Moisture content = 7.8 / 105.0 × 100 = **7.43%**.

Same water, two numbers: regain is always the larger, because its denominator is smaller. Trade and testing use regain; every value in this sheet is regain unless stated.

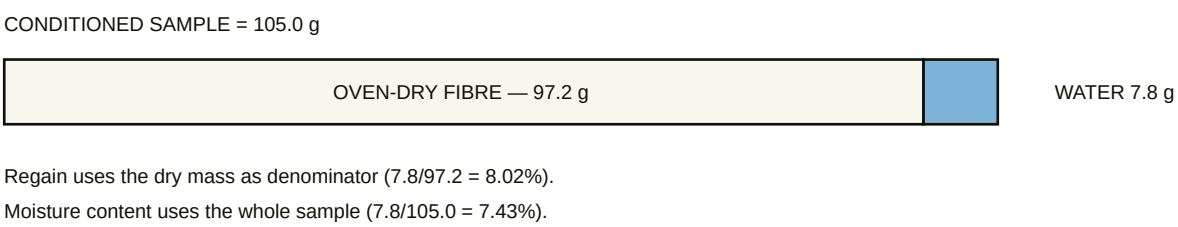


Figure 1 — The composition of the worked-example sample, to scale: 97.2 g of dry fibre carrying 7.8 g of water.

2. Standard regains

Because regain drifts with humidity, contracts fix a **standard (official) regain** for each fibre, at which weight is settled and counts are quoted.

FIBRE	STANDARD REGAIN (%)	FIBRE	STANDARD REGAIN (%)
Cotton	8.5	Viscose rayon	13.0
Wool	16.0	Silk	11.0
Flax / jute	12.0	Nylon	4.5
Acrylic	2.0	Polyester	0.4

3. Correcting a count to standard regain

Count is length per unit mass, so a drier-than-standard sample is lighter and tests *finer* (higher count) than it should. The corrected count is:

$$\text{Corrected count} = \text{observed count} \times (100 + \text{observed regain}) / (100 + \text{standard regain})$$

WORKED EXAMPLE — COUNT CORRECTION

A cotton yarn tests at Ne 28.4 when its regain is 6.5%. Standard regain for cotton = 8.5%.
Corrected count = 28.4 × (100 + 6.5)/(100 + 8.5) = 28.4 × 106.5/108.5 = **Ne 27.9**.
Check the direction: at standard regain the yarn would carry more water, weigh more per unit length, and so count coarser (lower Ne) — 27.9, not higher. A correction that moves the wrong way signals the formula has been inverted.

4. Invoice (correct conditioned) weight

A consignment is weighed as received, at whatever regain the journey gave it, then recalculated to the mass it would have at standard regain. Only that mass is invoiced.

$$\text{Invoice weight} = \text{delivered weight} \times (100 + \text{standard regain}) / (100 + \text{actual regain})$$

WORKED EXAMPLE — INVOICE WEIGHT OF A COTTON YARN CONSIGNMENT

Delivered weight = 5000 kg at a tested regain of 10.2%. Standard regain = 8.5%.
Invoice weight = 5000 × 108.5 / 110.2 = **4922.9 kg**.
The buyer does not pay for the extra 77.1 kg of water above the standard regain. If the consignment had arrived drier than standard, the same formula would have increased the invoice weight in the seller's favour — the rule is symmetric.

5. Count conversions

Three systems dominate: cotton count (Ne, indirect), tex (direct, grams per 1000 m) and denier (direct, grams per 9000 m); Nm (metric count) appears with wool and man-made fibres.

$$\text{tex} = 590.5 / \text{Ne} \quad \cdot \quad \text{denier} = \text{tex} \times 9 \quad \cdot \quad \text{Nm} = 1.693 \times \text{Ne}$$

WORKED EXAMPLE — CONVERTING NE 20

tex = 590.5 / 20 = **29.5 tex**.

denier = 29.5 × 9 = **266 denier**.

Nm = 1.693 × 20 = **33.9**.

Sanity check by direction: an indirect count of 20 is a medium yarn; 29.5 g per kilometre and 266 g per 9000 m describe the same yarn — and 29.5 × 9 = 265.7 ≈ 266 confirms the pair.

6. Count from a weighed lea

In the laboratory, count is found by winding a lea (skein) of 120 yards on a wrap reel, weighing it, and converting. A lea is 1/7 of a hank (840 yd), so:

$$\text{Ne} = (120/840) / (\text{lea mass in lb}) = 0.142857 / \text{lea mass (lb)}$$

WORKED EXAMPLE — LEA TEST

A lea of 120 yd weighs 3.31 g = 3.31/453.6 = 0.007297 lb.

Ne = 0.142857 / 0.007297 = **19.6**.

Routine count testing averages many leas (20 or more from different packages) and reports the mean with its CV% — the quantity controlled in YM-4101, Topic 8.

Glossary

Regain — water in a sample as a percentage of its oven-dry mass. **Moisture content** — water as a percentage of the total sample mass. **Conditioning** — bringing a sample to equilibrium in the standard atmosphere (65% RH, 20 °C) before testing. **Standard regain** — the contract regain at which weight and count are settled for a fibre. **Invoice weight** — the mass of a consignment recalculated to standard regain. **Lea (skein)** — 120 yards of yarn wound on a 1.5-yard wrap reel, the standard count-test specimen. **Direct / indirect system** — count expressed as mass per length (tex, denier) or length per mass (Ne, Nm).

Key points

• Regain divides water by dry mass; moisture content divides by total mass — regain is always larger (8.02% vs 7.43% in the example).
• Cotton's standard regain is 8.5%; polyester's is only 0.4% — moisture arithmetic matters most for natural fibres.
• Corrected count = observed count × $(100 + R_{obs})/(100 + R_{std})$ — Ne 28.4 at 6.5% becomes Ne 27.9 at standard.
• Invoice weight uses the same ratio on mass: 5000 kg at 10.2% regain invoices as 4922.9 kg.
• tex = 590.5/Ne; denier = 9 × tex; a 120-yd lea of 3.31 g tests Ne 19.6.

Practice

1. A sample weighs 86.4 g conditioned and 80.0 g oven-dry. Calculate its regain and its moisture content.
2. Wool yarn (standard regain 16%) tests Ne 24.0 (worsted count basis ignored — treat as a count) at a regain of 13.5%. Calculate the corrected count at standard regain.
3. Convert Ne 36 to tex and denier. A lea of this yarn should weigh how many grams?
4. A consignment of cotton yarn weighs 3200 kg at 7.6% regain. Calculate the invoice weight at the standard regain of 8.5%.

Answer key

1. Water = 6.4 g. Regain = 6.4/80.0 × 100 = **8.0%**. Moisture content = 6.4/86.4 × 100 = **7.41%**.
2. Corrected = 24.0 × 113.5/116.0 = **23.5**. (Drier than standard → observed count too fine → corrected count lower.)
3. tex = 590.5/36 = **16.4 tex**; denier = 16.4 × 9 = **147.6 ≈ 148**. Lea mass: Ne 36 → mass per lea = 0.142857/36 lb = 0.003968 lb = **1.80 g**.
4. Invoice = 3200 × 108.5/107.6 = **3226.8 kg**. (Drier than standard, so invoice weight exceeds delivered weight.)